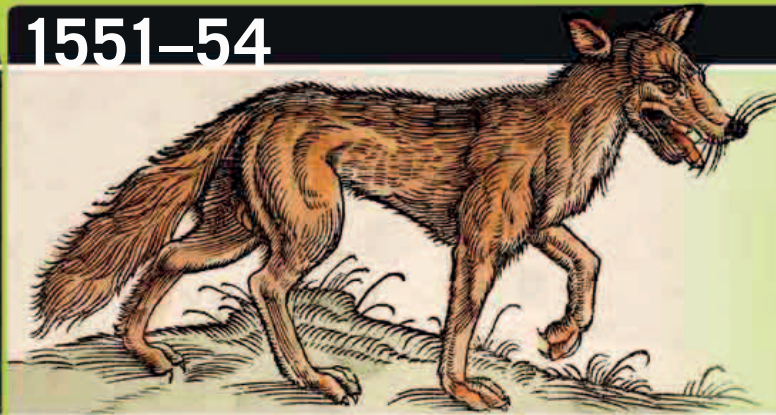




Fracastoro made major advances in understanding the spread of disease.



Konrad von Gesner published several volumes of *Historia Animalium*, illustrated with dramatic pictures and accurately detailed drawings.

“LET THERE BE FOR EVERY PULSE A THANKSGIVING, AND FOR EVERY BREATH A SONG.”

Konrad Von Gesner, German naturalist, 1550s

emerging study of geology. After examining the fossils of marine creatures found by building workers who were excavating a site in Verona, he expressed the **controversial idea** that they may be the **fossilized remains** of animals that had lived there many years before.

This, however, was not the view held by other geologists of the time. **German scholar** Georg Pawer, known as **Georgius Agricola** (1494–1555), dismissed the idea. He maintained that these were organic shapes created by the action of heat on “fatty matter” within the rocks. Despite this erroneous opinion, Agricola was among the first to lay a **scientific foundation** for the **study of geology**. In his 1546 publication *De Veteribus et Novis Metallis*, better known as *De Natura Fossilium*, he attempted to categorize various minerals and rocks according to their characteristics. This, along with his earlier text *De Re Metallica*, provided a comprehensive **overview of mineralogy and geology**, and was a practical guide to various mining techniques and machinery used at the time. It also showed the inadequacy of contemporary theories, which had not changed since the time of the Romans.

ENGLISH SURVEYOR LEONARD DIGGES (1520–59) made measurement of distance more accurate with the invention, in 1551, of the **theodolite**.

In the same year, **German naturalist Konrad von Gesner** (1516–65) published the first of five volumes of his *Historia*

Animalium (*The History of Animals*). This book attempted to present a comprehensive **catalog** of the real and mythical **creatures of the world** and included illustrations and engravings. More importantly, it introduced exotic and recently discovered animals to European

readers. Despite its popularity in northern Europe, the series was banned by the Catholic Church because of von Gesner's Protestant beliefs.

The anatomical drawings of Italian physician **Bartolomeo Eustachi** (c.1520–74), completed in 1552, were not published until 1714 because he feared excommunication from the Catholic Church. He studied human teeth and was the first to describe **adrenal glands**, but he is more commonly known for his research into the workings of the ear, specifically the tube now known as the **Eustachian tube**.

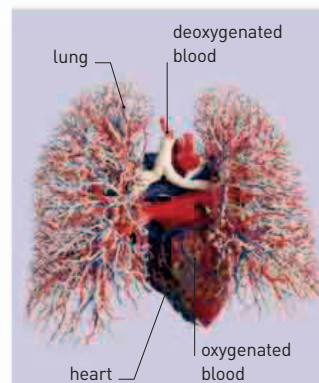
Michael Servetus (see panel, opposite) published his *Christianismi Restitutio* (*The Restoration of Christianity*) in 1553, but fell foul of both the Catholic and Protestant authorities in doing so. In it, he included the first correct description of **pulmonary circulation**.

Also controversial was the theory proposed by **Giambattista Benedetti** (1530–90) concerning “bodies” (objects) in free fall. In his book published in 1554, he stated that

Theodolite

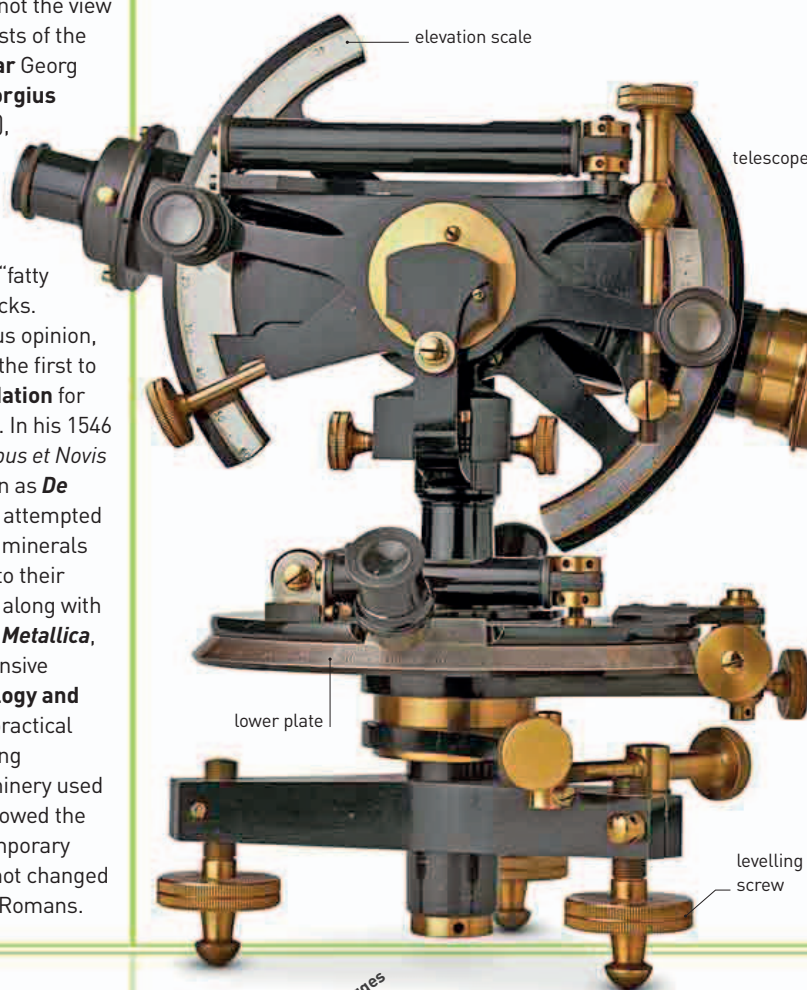
The theodolite is used to measure vertical and horizontal angles. This modern example is equipped with a telescope, which enables surveying over even longer distances.

bodies of the same material fell at the same speed, no matter what their weight, contradicting the law proposed by Aristotle. In a second edition of the work, he modified his theory to account for air resistance (friction), but maintained that different-sized bodies would fall at the same speed in a vacuum.



PULMONARY CIRCULATION

The heart pumps deoxygenated blood through the pulmonary artery to small capillaries around the lungs, where carbon dioxide is replaced with oxygen. The pulmonary vein returns this oxygen-rich blood to the heart. Michael Servetus was the first to describe this system in 1553, but it had little influence at the time.



1550 Gerolamo Cardano publishes a survey of the natural sciences, *De Subtilitate Rerum*

1551 Leonard Digges invents theodolite

1552 Bartolomeo Eustachi finishes his *Anatomical Engravings*

1554 Giambattista Benedetti proposes theory of bodies in free fall

1550s Taqi al-Din designs programmable weight-driven astronomical clock

1551 Konrad von Gesner publishes first volume of *Historia Animalium*

1553 Spanish physician Michael Servetus publishes *Christianismi Restitutio*